

Qun Chen, PhD

Associate Research Fellow

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RESEARCH INTERESTS

My research sits at the interface of clinical multi-omics, biomedical AI, and translational data science. I build interpretable machine-learning and statistical models that connect large-scale molecular signatures (plasma proteomics, metabolomics) to disease trajectories, clinical outcomes, and biological mechanisms, with a long-term goal of bridging population-level discovery and AI-guided biological design.

EDUCATION

- **Tsinghua University** Shenzhen, Guangdong, China
Ph.D. in Biomedical Engineering Sep. 2017 – Dec. 2022
 - **Dissertation:** Molecular detection technologies for rapid infectious disease diagnosis and biomarker discovery.
- **University of Chinese Academy of Sciences (UCAS)** Shanghai, China
M.Sc. in Bioengineering Sep. 2013 – Jan. 2017
- **Anhui Medical University** Hefei, Anhui, China
B.Sc. in Food Quality and Safety Sep. 2009 – Jun. 2013

POSITIONS

- **The First Affiliated Hospital of Xiamen University** Xiamen, China
Associate Research Fellow Aug. 2023 – Present
 - **Affiliation:** Xiamen Cell Therapy Research Center & Xiamen University School of Medicine.
 - **Research Focus:** Clinical multi-omics, cardiovascular disease trajectory modeling, medical AI.
- **University of California, Berkeley** Berkeley, CA, USA
Visiting Ph.D. Student Oct. 2018 – Sep. 2019
 - **Host:** Professor Ting Xu.

SELECTED PUBLICATIONS

(**Bold** = author; * co-first author; †corresponding author)

1. Su D, **Chen Q*** (co-first), Xu R, Chen Q, Ma C, Chen X, Yang Y, Qin F, Zhou Z, Li S, Peng J, Ge D, Fan J, Li X, Wang Y, Yang Y, Guo Y, Yang J†, Ning W† (2026). Development and validation of a machine learning-based diagnostic system for 22 pediatric respiratory pathogens: a large-scale multicenter study (Pathog-PDx; 134,500 hospitalized children; 42 EHR features; mean AUC 0.88). *npj Digital Medicine*. doi:10.1038/s41746-026-02818-9.
2. Yang J, Ning W, Xu R, Guo Y, He A, Fan J, Wang Y, Li X, **Chen Q**† (corresponding) (2026). Metabolomic Atlas of Cardiovascular Diseases: Mapping Shared and Specific Signatures (325 NMR metabolites; 244,567 UK Biobank participants; 87 CVD phenotypes). *JACC: Advances*, 5(5): 102742. doi:10.1016/j.jacadv.2026.102742.
3. Lei S, Ning W, Zhang H, **Chen Q*** (co-first), Xu L, Tian S, Li X, Liu W, Xiang M, Liu F, Wang G, Wang L, Wang Z (2026). Deep visual multi-omics reveals intra-tumor heterogeneity of gene expression and immunophenotype in colorectal cancer. *Science Bulletin*. doi:10.1016/j.scib.2026.04.047.
4. **Chen Q***, Gul I*, Liu C, et al. (2023). CRISPR Cas12-based field-deployable system for rapid detection of synthetic DNA sequence of the monkeypox virus genome. *Journal of Medical Virology*, 95(1): e28385.

5. **Chen Q***, Wang N*, Zhu M, et al. (2018). TiO₂ nanoparticles cause mitochondrial dysfunction, activate inflammatory responses, and attenuate phagocytosis in macrophages: a proteomic and metabolomic insight. *Redox Biology*, 15: 266–276.
6. He Q, **Chen Q***, Lian L, et al. (2024). Unraveling the influence of CRISPR/Cas13a reaction components on enhancing trans-cleavage activity for ultrasensitive on-chip RNA detection. *Microchimica Acta*, 191(8): 466–477.
7. Yang G, Chen H, **Chen Q***, et al. (2023). Injury-induced interleukin-1 alpha promotes Lgr5 hair follicle stem cells de novo regeneration via regenerative microenvironment. *Inflammation and Regeneration*, 43(1): 14.
8. **Chen Q***, Li F*, Zuo X, et al. (2022). Reversible domain closure modulates GlnBP ligand binding affinity. *PLOS ONE*, 17(4): e0263102.

PATENTS (SELECTED)

Issued patents reflecting methodological contributions in medical AI, biological age prediction, and multi-omics integration:

1. **Chen Q**, Ning W, Zhang S, et al. Method and Apparatus for Brain Age Prediction Based on Deep Learning and Metabolomics Data. CN202411303281.8 (2025).
2. Ning W, **Chen Q**, Zhang S, et al. Method and Apparatus for Cardiovascular Age Prediction Based on Deep Learning and Metabolic Profiling. CN202411384982.9 (2024).
3. Ning W, **Chen Q**, Hong G, et al. Clustering Method and Apparatus for Cardiovascular and Cerebrovascular Disease Mortality Populations Based on DNN and Consensus Clustering. CN202411453198.9 (2024).
4. Additional patents in medical AI memory governance and pathology-image-to-molecular-state prediction (details available on request).

RESEARCH FUNDING

- **Fujian Provincial Natural Science Foundation** Grant No. 2025J08313
Principal Investigator (PI) *May 2025 – Apr. 2028*
 - **Project:** Study on Maternal Chronic Stress Inducing Neural Precursor Cell Differentiation Disorders via the Ncam1–MAPK Signaling Pathway.

TECHNICAL SKILLS

- **Multi-omics analysis:** plasma proteomics (UKB-PPP ~2,900 proteins), metabolomics, scRNA-seq, spatial omics.
- **Machine learning / AI:** XGBoost, Transformer, deep learning (PyTorch/TensorFlow), survival analysis (Cox, landmark), Mendelian randomization, mediation analysis.
- **Clinical data science:** longitudinal EHR analysis, disease trajectory modeling, real-world clinical data integration.
- **Molecular biology:** CRISPR-Cas12/13 diagnostics, protein biochemistry, nanomaterial biosafety (prior expertise).
- **Programming:** Python (primary), R, statistical computing.